

NAGOYA AP, Japan

WMO: 476350

Lat: 35.255N

Lon: 136.924E

Elev: 52

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.4	30.2	13.9	11.1	41.2	16.0	12.4	40.7	24.5	44.7	22.6	44.0	4.5	340	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.4	95.4	77.5	93.3	77.1	91.3	76.3	79.9	89.4	79.0	88.0	78.1	86.8	9.2	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	142.3	83.7	76.8	140.1	83.5	75.4	133.5	82.6	43.5	89.9	42.5	87.9	41.6	86.5	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.2	19.3	16.7	DB	24.5	99.3	2.3	2.2	22.9	100.9	21.5	102.2	20.2	103.4	18.5	105.0	
			WB	22.4	81.5	1.8	1.2	21.1	82.4	20.0	83.1	19.0	83.8	17.8	84.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	40.2	41.9	48.4	58.4	67.7	74.5	82.0	84.4	77.2	66.4	54.8	44.8
	DBStd	15.92	3.96	4.98	5.54	5.81	4.38	4.14	4.48	3.74	5.05	5.14	5.54	5.12
	HDD50	833	306	231	98	7	0	0	0	0	0	0	17	174
	HDD65	3135	770	645	514	206	23	0	0	1	43	308	625	
	CDD50	5154	0	6	50	260	549	734	992	1065	815	509	160	14
	CDD65	1980	0	0	0	9	107	284	527	600	365	87	1	0
	CDH74	18819	0	0	1	62	635	1858	5695	7150	3039	377	2	0
	CDH80	7430	0	0	0	5	121	507	2432	3333	989	43	0	0
Wind	WSAvg	7.0	7.1	7.9	8.3	7.9	7.3	6.4	6.6	6.8	6.6	6.3	6.1	6.6
Precipitation	PrecAvg	61.4	1.9	2.5	4.4	5.4	6.1	7.8	8.1	5.7	8.8	5.2	3.1	1.9
	PrecMax	80.0	6.1	6.2	8.3	9.1	13.0	20.2	17.7	14.9	27.9	20.9	9.1	4.9
	PrecMin	35.6	0.0	0.4	0.1	0.3	2.0	2.3	1.1	0.2	1.2	0.4	0.3	0.0
	PrecStd	10.2	1.3	1.5	1.9	2.0	2.5	3.1	4.1	3.7	5.1	3.6	1.9	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	62.7	69.9	79.2	87.8	91.7	98.6	98.9	93.5	84.1	73.3	64.0
		MCWB	45.7	51.9	53.7	59.3	64.9	74.0	78.1	78.0	76.7	69.7	60.9	54.8
	2%	DB	53.4	57.4	65.9	75.3	82.8	87.9	95.3	96.7	91.0	80.7	69.9	59.2
		MCWB	44.3	46.9	52.3	59.4	63.7	71.4	77.4	77.8	75.0	67.6	58.1	50.1
	5%	DB	50.4	54.0	62.5	71.9	80.4	85.6	93.1	94.7	88.0	78.5	67.7	56.9
		MCWB	42.5	45.0	51.5	57.5	63.3	70.6	77.1	77.5	74.0	66.2	57.4	48.8
	10%	DB	48.3	51.7	59.1	69.5	77.3	82.6	91.0	91.8	85.8	75.4	64.5	53.8
		MCWB	40.8	43.3	48.8	57.2	62.4	69.1	76.6	76.8	72.9	64.4	55.7	46.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.0	54.0	59.5	67.9	71.9	77.8	80.9	81.5	79.6	74.7	64.2	57.0
		MCDB	52.4	60.5	62.8	70.7	77.1	84.7	91.8	92.1	87.3	78.2	68.2	59.7
	2%	WB	45.9	50.2	56.3	63.8	69.1	76.1	79.7	80.2	77.9	71.5	61.7	52.8
		MCDB	49.8	54.9	61.9	69.3	75.0	82.7	89.5	89.7	84.6	77.4	66.5	57.2
	5%	WB	43.7	46.9	53.5	61.6	66.9	74.4	78.7	79.4	76.7	68.9	59.6	49.9
		MCDB	48.5	51.7	60.3	68.5	74.3	80.8	88.0	88.7	83.6	74.7	64.7	54.6
	10%	WB	41.7	44.0	50.6	59.3	65.2	72.4	77.8	78.6	75.5	66.7	57.5	47.5
		MCDB	47.1	50.2	57.2	66.0	73.5	79.2	86.7	87.6	82.7	73.0	62.8	53.0
Mean Daily Temperature Range	5% DB	MDBR	14.7	15.9	17.1	17.8	16.9	13.9	13.5	14.4	13.9	15.1	15.7	14.7
		MCDBR	17.6	19.4	21.4	22.0	22.2	18.0	17.0	17.2	16.0	17.0	18.0	17.4
		MCWBR	11.4	12.2	12.4	10.6	9.0	6.5	5.5	5.2	5.8	7.9	10.1	10.9
	5% WB	MCDBR	15.1	15.9	16.9	15.4	15.2	12.7	14.7	14.6	12.8	13.0	13.7	14.4
		MCWBR	11.2	12.1	13.1	10.2	9.1	6.0	5.5	5.5	5.8	8.1	9.8	11.0
Clear-Sky Solar Irradiance	taub	0.391	0.430	0.497	0.540	0.549	0.578	0.567	0.545	0.496	0.458	0.415	0.390	
	taud	2.163	2.070	1.920	1.867	1.914	1.913	1.996	2.045	2.149	2.173	2.200	2.189	
	Ebn at Noon	248	252	244	241	241	233	235	238	243	240	238	239	
	Edh at Noon	37	45	57	63	61	61	56	52	45	41	36	34	
All-Sky Solar Radiation	RadAvg	741	965	1264	1513	1661	1501	1549	1594	1271	1038	797	656	
	RadStd	52	72	90	144	181	135	218	148	115	101	68	54	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+193	+124
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page